

## Introduction

The negative binomial distribution is the workhorse model for overdispersed counts – counts whose variance exceeds their mean. It arises naturally as the number of failures before the  $r$ -th success in a sequence of Bernoulli trials, or as a gamma mixture of Poisson distributions. It dominates modern count-regression practice (RNA-seq, public-health incidence, insurance claims).

## Prerequisites

Geometric and Poisson distributions; gamma distribution is useful for the mixture interpretation.

## Theory

Two standard parameterisations:

**Counts-of-failures form:**  $Y$  is the number of failures before the  $r$ -th success in iid Bernoulli( $p$ ) trials:

$$P(Y = k) = \binom{k+r-1}{k} p^r (1-p)^k, \quad k = 0, 1, 2, \dots$$

**Mean-dispersion form:** parameterise by mean  $\mu$  and dispersion (size) parameter  $\theta$  with

$$P(Y = k) = \frac{\Gamma(k+\theta)}{k! \Gamma(\theta)} \left( \frac{\theta}{\theta + \mu} \right)^\theta \left( \frac{\mu}{\theta + \mu} \right)^k.$$

Moments in the mean-dispersion form:

- $E[Y] = \mu$ .
- $\text{Var}(Y) = \mu + \mu^2/\theta$ .

**Gamma-Poisson interpretation:** if  $\Lambda \sim \text{Gamma}(\theta, \theta/\mu)$  and  $Y \mid \Lambda \sim \text{Poisson}(\Lambda)$ , then marginally  $Y \sim \text{NegBin}(\mu, \theta)$ . As  $\theta \rightarrow \infty$ , the NB collapses to  $\text{Poisson}(\mu)$ .

**Geometric as a special case:**  $r = 1$  (or  $\theta = 1$ ) recovers the geometric distribution.

## Assumptions

No common-rate assumption (unlike Poisson); the heterogeneity in underlying rates is parameterised by  $\theta$ .

## R Implementation

```
library(MASS)

# R's rnbinom: parameterise by size (= theta) and mu
mu <- 5; theta <- 2
Y <- rnbinom(1e5, size = theta, mu = mu)

c(mean      = mean(Y),
  variance = var(Y),
  expected_var = mu + mu^2 / theta)

# PMF comparison to Poisson
k <- 0:20
pmf_nb  <- dnbinom(k, size = theta, mu = mu)
pmf_pois <- dpois(k, lambda = mu)
```

```
data.frame(k, nb = round(pmf_nb, 4), poisson = round(pmf_pois, 4))

# Fit by MLE using MASS::glm.nb
df <- data.frame(y = Y, x = rnorm(1e5))
fit <- glm.nb(y ~ x, data = df)
summary(fit)$theta
```

## Output & Results

```
      mean  variance expected_var
5.020    17.48      17.50

      k      nb  poisson
1  0  0.2041  0.0067
2  1  0.2041  0.0337
3  2  0.1701  0.0842
4  3  0.1296  0.1404
5  4  0.0925  0.1755
6  5  0.0634  0.1755
...

[1] 1.998      # estimated theta
```

Empirical variance (17.5) matches the theoretical  $\mu + \mu^2/\theta = 5 + 25/2 = 17.5$ . The MLE of  $\theta$  is close to the true value 2.

## Interpretation

Negative binomial is the default count-data model in modern applied work. In a manuscript: “incidence rates were compared using negative binomial regression to account for overdispersion (dispersion estimated as  $\theta = 2.0$ )”. Always report the dispersion parameter alongside the rate ratios.

## Practical Tips

- Detect overdispersion by fitting a Poisson first and checking residual deviance  $\approx$  df; ratios  $> 1.5$  suggest overdispersion.
- `glm.nb` jointly estimates regression coefficients and dispersion; faster-but-less-flexible alternatives include `glmmTMB` and `brms`.
- Zero-inflated variants (`pscl::zeroinfl` with `dist = "negbin"`) handle excess zeros on top of overdispersion.
- Reparameterisations: some textbooks use  $(n, p)$ , some  $(\mu, \theta)$ ; R’s `rnbinom(n, size, prob)` or `rnbinom(n, size, mu)` accepts either.
- In RNA-seq, DESeq2 and edgeR both model counts as negative binomial with shrinkage on the dispersion parameter.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

### See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots